JAVA LAB WEEK 9

Title: Packages – Defining, Creating and Accessing a Package, Importing Packages (Roman to Integer Conversion)

1. Objective / Aim

To write and execute a Java program to demonstrate the creation and use of a package named number, define a Roman class inside the package, and implement the romanToInteger() method to convert a Roman numeral into its integer value.

2. System Requirements

Hardware Requirements:

- Computer or laptop
- Minimum 4 GB RAM
- Keyboard and monitor

Software Requirements:

- Operating System: Windows, Linux, or macOS
- Java Development Kit (JDK): JDK 8 or above
- IDE: Eclipse, IntelliJ IDEA, NetBeans, or Visual Studio Code
- Command Prompt or Terminal

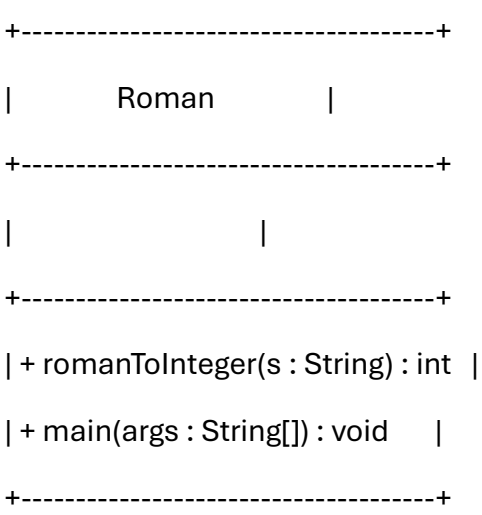
3. Algorithm / Logic Description

1. Start the program.
2. Create a package named number.
3. Define a public class named Roman inside the package.
4. Import the Scanner class to accept input from the user.
5. Define the romanToInteger() method that accepts a Roman numeral string.
6. Initialize an integer variable result to zero.
7. Read each character of the Roman numeral using a loop.
8. Use a switch statement to identify the Roman numeral character.
9. Add the corresponding integer value to result.
10. Return the final integer value from the method.
11. In the main() method, read a Roman numeral from the user.

12. Call the romanToInteger() method and display the converted integer value.
13. Stop the program.

4. UML Class Diagram

Package: number



Description:

- Roman is a public class inside the number package.
- romanToInteger(String s) is a static method that converts a Roman numeral into an integer.
- main(String[] args) is the entry point of the program.
- Scanner is used to read input from the user.
- No inheritance or object creation is required because the conversion method is static.

5. Source Code

```
package number;

import java.util.Scanner;

public class Roman {

   public static int romanToInteger(String s) {
```

```
    int result = 0;

    for (int i = 0; i < s.length(); i++) {
        switch (s.charAt(i)) {
            case 'I': result += 1; break;
            case 'V': result += 5; break;
            case 'X': result += 10; break;
            case 'L': result += 50; break;
            case 'C': result += 100; break;
            case 'D': result += 500; break;
            case 'M': result += 1000; break;
        }
    }

    return result;
}

public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    System.out.print("Enter Roman Number: ");
    String roman = sc.nextLine();

    int result = romanToInteger(roman);

    System.out.println("Integer Value: " + result);

    sc.close();
```

```
    }
}
```

6. Compilation and Execution Commands

Step 1: Create the folder structure

Create a folder named Project. Inside it, create a folder named number and save Roman.java inside the number folder.

```
Project
  |
  +-- number
      |
      +-- Roman.java
```

Step 2: Open Command Prompt or Terminal

Navigate to the Project folder.

Step 3: Compile the program

```
javac -d . number/Roman.java
```

Step 4: Execute the program

```
java number.Roman
```

Note: Since the class belongs to the number package, use java number.Roman to execute it.

7. Output Screen

Sample Output:

```
Enter Roman Number: LVIII
Integer Value: 58
```

Explanation of Output:

The input Roman numeral is LVIII.

- L = 50
- V = 5
- I = 1
- I = 1

- I = 1

Total value:

50 + 5 + 1 + 1 + 1 = 58

Therefore, the output is 58.

8. Result

Thus, the Java program to demonstrate package creation and the Roman numeral conversion method was successfully implemented and executed. The experiment demonstrates package definition, class creation, static methods, importing Java library classes, user input using Scanner, and the use of a switch statement for Roman numeral conversion.

Note: This program assumes that the input is a valid Roman numeral in standard order. It does not validate invalid Roman numeral combinations.